

PROJECT REPORT

Serverless Attendance System with Face Recognition

A Real-Time, Contactless Employee Attendance System on AWS

Prepared by:

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Cloud Platform: Amazon Web Services (AWS)

Core Services: API Gateway · Lambda · S3 · Rekognition · DynamoDB · EventBridge · IAM · CloudWatch

1. Abstract

This report presents the design and implementation of the **Serverless Employee Face Recognition and Attendance Management System**, a cloud-based solution developed using Amazon Web Services (AWS). The system enables administrators to register employees by storing their employee details and reference face images, while employees can capture attendance images through a web-based application. The captured images are stored temporarily in Amazon S3 and processed using Amazon Rekognition for face comparison. When a registered employee is successfully identified, the attendance record is stored in Amazon DynamoDB. AWS Lambda functions handle employee registration, image processing, attendance management, dashboard operations, archival, and image cleanup. Amazon EventBridge is used to automate scheduled attendance archival. The system uses a serverless, event-driven architecture to reduce infrastructure management and provide a scalable approach to employee attendance management.

2. Introduction

Traditional attendance systems rely on manual registers, biometric fingerprint devices, or RFID/ID-card punch systems. These approaches typically involve physical contact, dedicated hardware, on-premise infrastructure and manual reconciliation of records. This project addresses these limitations by using facial recognition as a contactless, automated method of identifying employees, combined with a fully serverless AWS architecture so that no servers need to be provisioned, patched, or scaled manually.

The system captures an employee's face using a standard webcam or mobile browser camera, uploads the image securely to cloud storage and uses Amazon Rekognition to match it against a pre-registered collection of employee faces. Once identified, the system records the employee's attendance information in Amazon DynamoDB. Scheduled processing is used to archive daily attendance records for administrative access.

3. Objectives

- Design and implement a contactless employee attendance system using facial recognition.
- Build the application using a serverless AWS architecture.
- Provide an administrator interface for registering employees and managing attendance information.
- Store registered employee details and reference face images securely using AWS services.
- Automate face comparison using Amazon Rekognition.
- Record successful attendance events in Amazon DynamoDB.
- Automate daily attendance archival using Amazon EventBridge and AWS Lambda.
- Provide separate employee and administrator web interfaces.
- Apply security practices such as authentication, least-privilege IAM permissions, encryption, and temporary storage of attendance images.

4. System Scope

The system covers employee registration, employee face-image storage, attendance image capture, facial recognition, attendance recording, administrative dashboard access, daily attendance archival, and temporary image cleanup. The solution uses AWS managed services including Amazon Cognito, API Gateway, Lambda, S3, Rekognition, DynamoDB, EventBridge, IAM, and CloudWatch.

5. System Architecture

The system follows a serverless and event-driven AWS architecture consisting of employee registration, attendance processing, administrative dashboard, and scheduled archival workflows. Administrators authenticate using Amazon Cognito and can register employees through the Employee Registration API. Employee details are stored in Amazon DynamoDB and reference face images are stored in Amazon S3.

During attendance, employees use the employee web application to capture and upload a face image through the Get Image API. The image is temporarily stored in Amazon S3. The S3 object-created event triggers the Image Recognition Lambda, which uses Amazon Rekognition to compare the captured image with registered employee reference images. When a face is successfully matched, the Attendance Employee Lambda records the attendance information in the Emp-Attendance DynamoDB table.

Amazon EventBridge periodically triggers the Archive Attendance Lambda to read attendance records and store daily attendance archives in Amazon S3. A separate Lambda function is responsible for deleting temporary attendance images after processing.

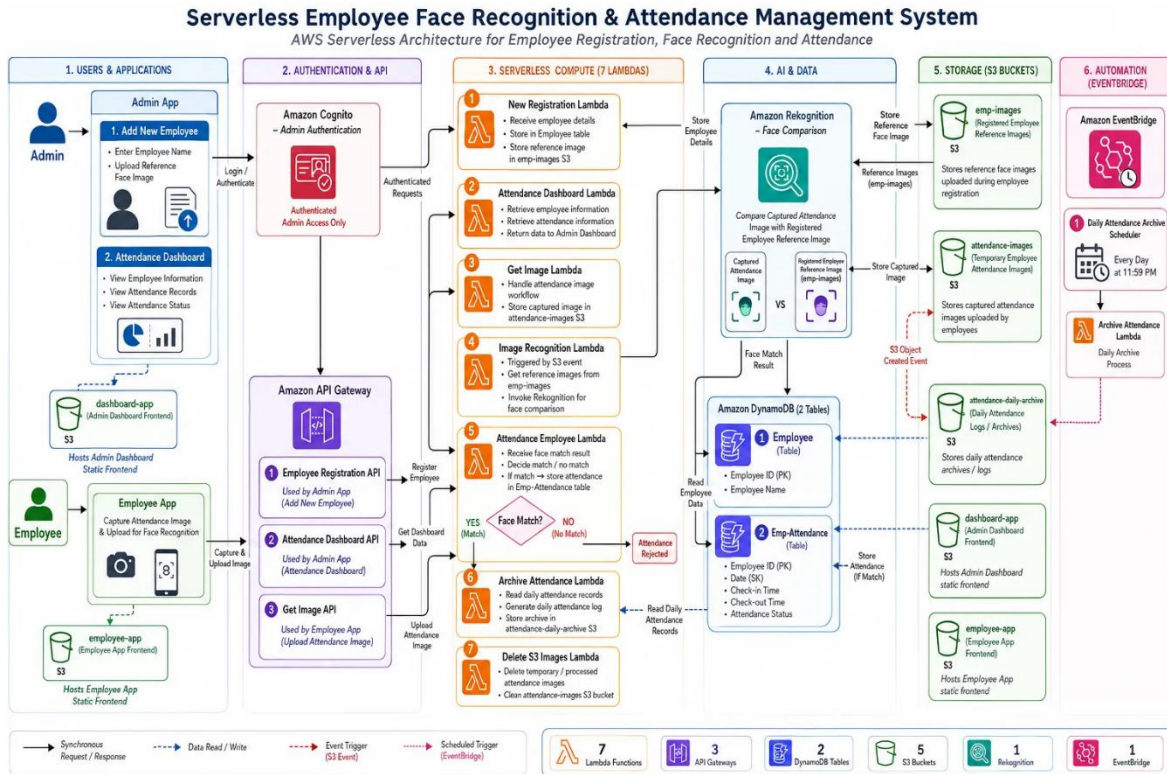


Figure 1: End-to-end system architecture

High-Level Workflow

Employee Registration

- The administrator opens the Admin Application and authenticates using **Amazon Cognito**.
- The administrator enters employee details and uploads a reference face image.
- The Employee Registration API receives the registration request through **Amazon API Gateway**.
- The **New Registration Lambda** processes the employee details.
- Employee information is stored in the **Employee DynamoDB table**.
- The employee's reference face image is stored in the **emp-images S3 bucket**.

Attendance Processing

- The employee opens the Employee Application and captures an attendance image.
- The image is sent through the **Get Image API** using Amazon API Gateway.
- The **Get Image Lambda** handles the image-upload workflow and stores the captured image in the **attendance-images S3 bucket**.
- The S3 object-created event triggers the **Image Recognition Lambda**.
- The Image Recognition Lambda obtains the registered reference image and invokes **Amazon Rekognition** for face comparison.
- Rekognition returns the face-match result.
- The **Attendance Employee Lambda** processes the result.
- If the face is successfully matched, the attendance information is stored in the **Emp-Attendance DynamoDB table**.
- If no valid match is found, the attendance request is rejected.
- **Amazon EventBridge** triggers the Archive Attendance Lambda according to the configured schedule.
- The Archive Attendance Lambda reads attendance records from DynamoDB.
- The daily attendance records are archived in the **attendance-daily-archive S3 bucket**.
- The Delete S3 Images Lambda removes temporary attendance images from the attendance-images bucket according to the implemented cleanup process.
- The administrator can view employee and attendance information through the **Attendance Dashboard**.

6. AWS Services Used

Service	Purpose in This Project
Amazon S3	Hosts the Web UI and Admin Dashboard; stores captured images and daily/weekly reports.
Amazon API Gateway	Exposes the HTTP(S) endpoint the frontend uses to request a pre-signed upload URL.
AWS Lambda	Runs all backend logic — seven single-purpose functions, with no dedicated servers.
Amazon Rekognition	Performs facial recognition by searching the captured image against the registered face collection.
Amazon DynamoDB	Stores attendance records (employee_id, date, clock_in, clock_out, status).
Amazon EventBridge	Triggers scheduled Lambdas for daily absence marking and reporting.
AWS IAM	Provides least-privilege roles/permissions scoped to each Lambda's actual needs.
Amazon CloudWatch	Provides logging, monitoring, and alerting across all Lambda executions.
Amazon Cognito	Provides authentication for administrators, ensuring only authorized users can access the Admin Dashboard.

7. Frontend Design

The system provides two web-based interfaces: an **Employee Application** for attendance capture and an **Admin Dashboard** for employee and attendance management. Both applications are hosted as static web applications using Amazon S3.

1. Employee Application

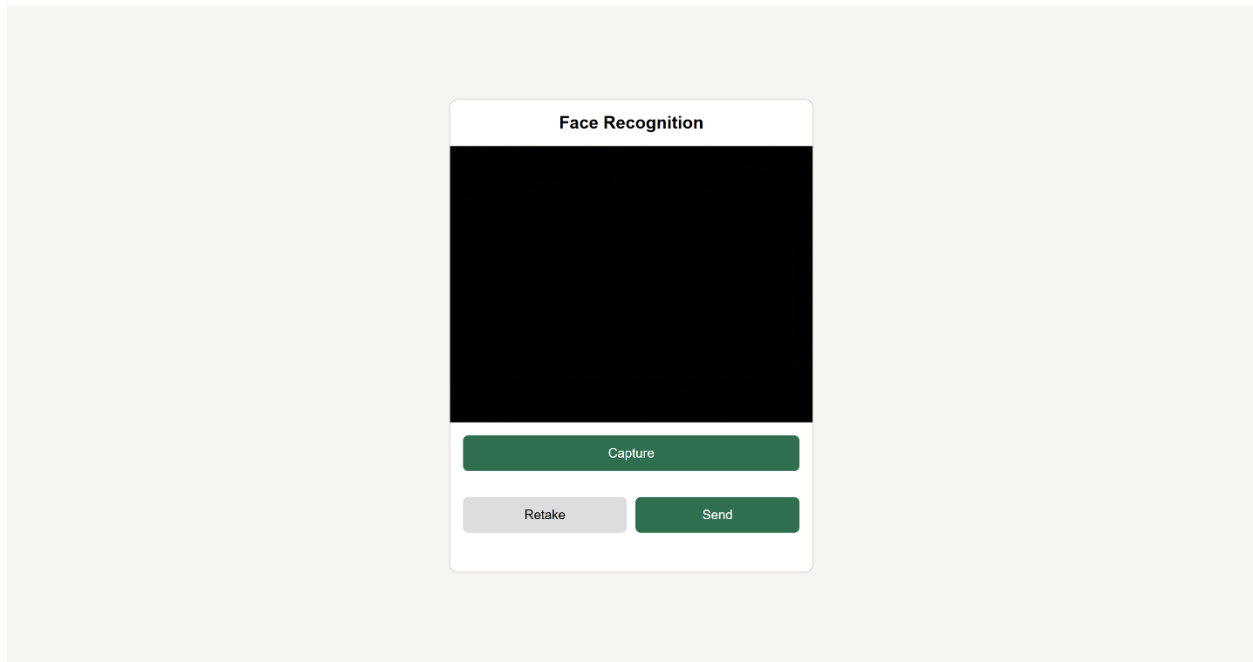
The Employee Application provides a simple interface for employees to capture their face image for attendance.

Face Recognition Interface

The interface provides:

- A live camera preview for capturing the employee's face.
- A **Capture** button to take the image.
- A **Retake** option to capture the image again if required.

A **Send** button to submit the captured image for face recognition and attendance processing.



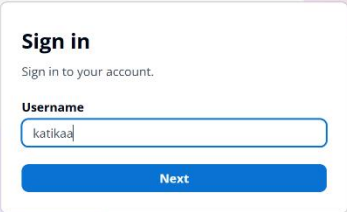
2. Admin Authentication

The Admin application provides a sign-in interface for authorized administrators.

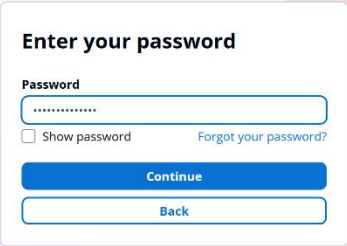
The authentication interface consists of:

- Username entry.
- Password entry.
- Continue/Back navigation.
- Password visibility option.
- Forgot-password functionality.

Authentication is handled using **Amazon Cognito**, ensuring that only authenticated administrators can access the attendance management dashboard.



The image shows a 'Sign in' form centered on a background with abstract purple, yellow, and grey shapes. The form has a title 'Sign in' and a subtitle 'Sign in to your account.' Below this is a 'Username' label and a text input field containing 'katikaaj'. At the bottom of the form is a blue button labeled 'Next'.



The image shows an 'Enter your password' form centered on the same abstract background. The form has a title 'Enter your password' and a 'Password' label. Below the label is a password input field with masked characters. To the left of the input field is a checkbox labeled 'Show password'. To the right is a link labeled 'Forgot your password?'. At the bottom of the form are two buttons: a blue 'Continue' button and a white 'Back' button.

3. Admin Attendance Dashboard

After successful authentication, the administrator is presented with the **Attendance Dashboard**.

The dashboard provides an overview of employee attendance, including:

- **Total Employees**
- **Present employees**

- **Late employees**
- **Absent employees**
- Attendance percentage overview.
- Detailed attendance records.

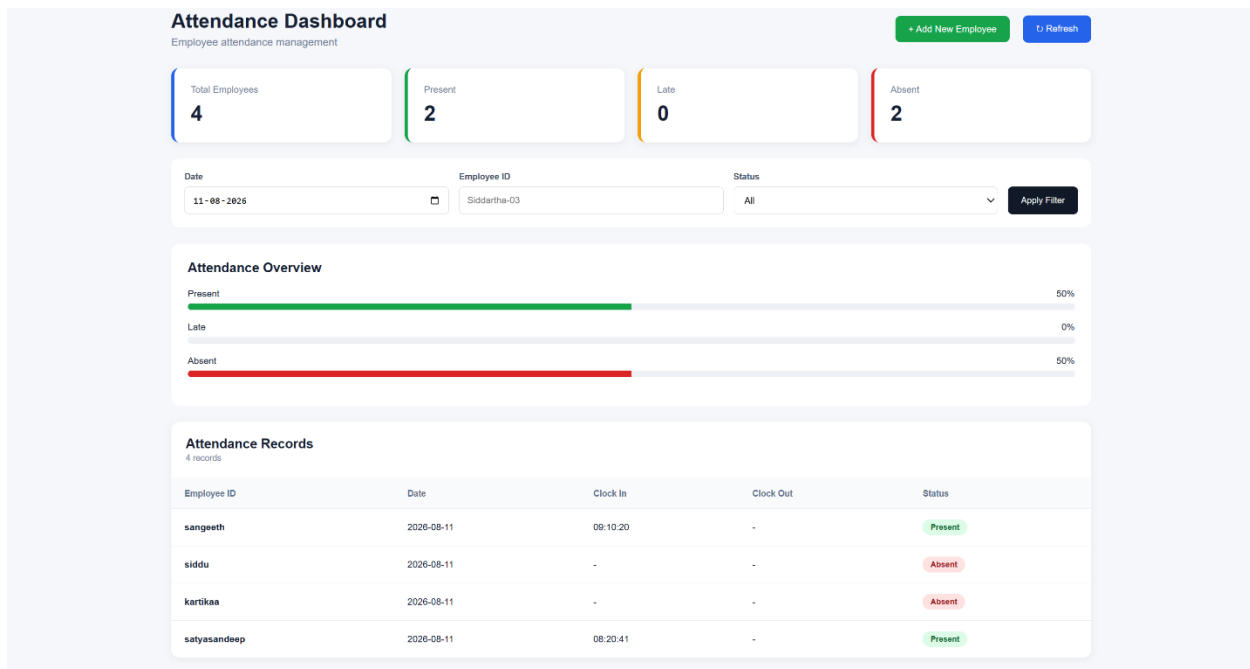
The attendance records display:

Field	Description
Employee ID	Identifies the employee
Date	Attendance date
Clock In	Employee check-in time
Clock Out	Employee check-out time
Status	Present/Absent/Late status

The dashboard also provides filtering options based on:

- Date
- Employee ID
- Attendance status

An **Add New Employee** option is also provided for employee registration, along with a **Refresh** option to retrieve the latest attendance information.



4. Frontend-to-Backend Integration

The frontend applications communicate with the AWS backend through **Amazon API Gateway**.

The major interactions are:

Employee Application

Employee → Employee App → Get Image API → Get Image Lambda → S3 → Face Recognition Pipeline

Admin Application

Admin → Cognito Authentication → Admin Dashboard → Attendance Dashboard API → Attendance Dashboard Lambda → DynamoDB

This separation allows employees to perform attendance capture while administrators can securely manage employees and view attendance information.

8. Lambda Functions

The system uses seven AWS Lambda functions, each responsible for a specific backend operation.

Lambda Function	Purpose
New Registration Lambda	Processes new employee registration, stores employee information in DynamoDB, and stores the reference face image in the emp-images S3 bucket.
Attendance Dashboard Lambda	Retrieves employee and attendance information required by the administrator dashboard.
Get Image Lambda	Handles the attendance image upload workflow and stores captured employee images in the attendance-images S3 bucket.
Image Recognition Lambda	Triggered when an attendance image is uploaded to S3. It retrieves the required reference image and invokes Amazon Rekognition for face comparison.
Attendance Employee Lambda	Processes the face-match result and records attendance when a valid employee match is obtained.
Archive Attendance Lambda	Runs on the scheduled EventBridge trigger, reads attendance records, and creates daily attendance archives in S3.
Delete S3 Images Lambda	Removes temporary or processed attendance images from the attendance-images S3 bucket.

9. Amazon Rekognition

During employee registration, the employee's reference face image is stored in the emp-images S3 bucket. During attendance processing, the captured attendance image is compared with the registered employee reference image using Amazon Rekognition. The resulting face-match information is passed to the Attendance Employee Lambda, which determines whether attendance should be recorded.

10. Amazon DynamoDB Design

The system uses two DynamoDB tables. The Employee table stores registered employee information captured during registration, such as employee ID, name, and other profile details. The Emp-Attendance table stores attendance records, using a composite primary key of `employee_id` (partition key) and `date` (sort key). This supports efficient lookup of all records for a given employee, and of a specific employee's record on a specific date. The attributes of the Emp-Attendance table are as follows:

Attribute	Key Type	Description
<code>employee_id</code>	Partition Key	Unique identifier of the employee
<code>date</code>	Sort Key	Date of the attendance record (YYYY-MM-DD)
<code>clock_in</code>	Attribute	Timestamp when the employee clocked in
<code>clock_out</code>	Attribute	Timestamp when the employee clocked out
<code>status</code>	Attribute	Attendance status: present, late, or absent

DynamoDB is well suited here: it is fully managed and serverless, scales automatically with read/write demand, offers single-digit-millisecond latency for real-time writes from the recognition pipeline, and the `employee_id` + `date` key naturally models per-employee, per-day records with efficient queries.

11. Attendance Logic and Reporting

When an employee's captured face image is successfully matched with a registered employee, the Attendance Employee Lambda processes the match result and records the attendance information in the Emp-Attendance DynamoDB table. The table maintains employee ID, date, check-in time, check-out time, and attendance status.

If the face cannot be successfully matched, the attendance request is rejected and no attendance record is created. For archival purposes, Amazon EventBridge triggers the Archive Attendance Lambda according to the configured schedule. The Lambda reads the attendance records and generates a daily attendance archive, which is stored in the `attendance-daily-archive` S3 bucket.

12. Security and Compliance

- IAM least privilege — each Lambda has a dedicated execution role scoped only to the specific resources and actions it needs (e.g., the Face Recognition Lambda can read the image bucket and call Rekognition, but cannot write to DynamoDB).
- S3 lifecycle deletion — the “attendance-images” bucket automatically deletes raw captured face images one hour after upload; only the derived outcome (matched ID, timestamp, status) is retained long-term in DynamoDB, minimizing exposure of biometric data.
- DynamoDB encryption at rest via AWS KMS.

- S3 versioning on relevant buckets, protecting reports against accidental overwrite or deletion.
- CloudWatch monitoring and logging across all Lambda functions for execution tracing and alerting.

13. Error Handling

Key failure scenarios and how the system is expected to respond:

- Camera permission denied — the frontend detects this and asks the employee to enable camera access.
- No face detected / invalid image — Rekognition returns no match; the system shows a “no face detected” message and records no attendance.
- Upload failure — the frontend detects the failed S3 upload and prompts the employee to retry.
- Low-confidence or no match — the match is rejected below the configured threshold; the employee is treated as unrecognized and attendance is not recorded.
- Lambda or DynamoDB failure — the error is logged to CloudWatch for troubleshooting, and API Gateway returns an error response to the frontend where applicable.

14. Testing and Verification

The following table consolidates the test cases used to verify the end-to-end functionality of the system, from image capture through attendance recording and reporting.

Test Case	Test Description	Expected Result
Camera access	Verify the employee web application requests and obtains camera permission before capture.	Camera preview displays; if permission is denied, the app prompts the employee to enable access.
Image capture	Verify the Capture and Retake controls correctly capture and allow re-capture of a face image.	A usable image is captured and displayed for review before submission.
Upload URL generation	Verify the Get Image API returns a valid pre-signed upload URL via API Gateway.	A valid, time-limited pre-signed URL is returned to the frontend.
S3 upload	Verify the captured image uploads successfully to the attendance-images S3 bucket using the pre-signed URL.	Image object is created in the attendance-images bucket; upload failures prompt a retry.
S3 event trigger	Verify that an S3 object-created event correctly triggers the Image Recognition Lambda.	Image Recognition Lambda is invoked automatically upon upload.
Face recognition	Verify Amazon Rekognition compares the captured image against the registered employee reference collection.	Rekognition returns a face-match result with a confidence score.

Test Case	Test Description	Expected Result
Correct employee ID	Verify that a successful match returns the correct, corresponding employee ID.	The employee ID returned matches the registered employee in the reference image.
Low-confidence / no-match	Verify behavior when Rekognition returns no match or a match below the configured confidence threshold.	Attendance request is rejected; employee is treated as unrecognized and no record is created.
DynamoDB attendance write	Verify the Attendance Employee Lambda writes a valid attendance record to the Emp-Attendance table on a successful match.	A record with employee_id, date, clock_in/clock_out, and status is written correctly.
Daily absence processing	Verify the scheduled process correctly identifies employees with no attendance record for the day.	Employees without a clock-in are correctly marked absent for that date.
Daily report generation	Verify the Archive Attendance Lambda generates and stores a daily attendance archive.	Daily archive file is created in the attendance-daily-archive S3 bucket on schedule.
Weekly analytics	Verify weekly attendance summaries can be derived correctly from the archived daily records.	Weekly totals and trends match the underlying daily attendance data.
Admin dashboard	Verify the Attendance Dashboard displays accurate, filterable employee and attendance information after Cognito authentication.	Dashboard shows correct totals (present/late/absent) and supports filtering by date, employee ID, and status.

15. AWS Cost Considerations

Cost is driven by usage across seven services: Rekognition (per image processed for search — at 100 employees and 2 recognition calls/day, roughly 6,000 calls/month), Lambda (invocations and GB-seconds across seven functions), API Gateway (per request), S3 (storage and PUT/GET requests), DynamoDB (read/write capacity and storage), EventBridge (per scheduled rule invocation), and CloudWatch (log ingestion and any alarms). Exact figures vary by region and change over time, so detailed pricing is provided as a separate cost sheet (a required project deliverable) rather than reproduced in full here.

16. Advantages

- Contactless attendance — reduces physical contact compared to fingerprint or card-based systems.
- High degree of automation — marking, absence detection, and reporting all happen without manual intervention.
- Serverless architecture — no servers to provision, patch, or manage.
- Scalability — AWS managed services scale automatically with employees and requests.
- Reduced manual work and centralized, dashboard-accessible attendance records.

17. Limitations

- Recognition accuracy can be affected by lighting, camera quality, and face pose/angle.
- Requires camera access, which may not be available or permitted on all devices.
- Requires continuous internet connectivity, since all processing runs on AWS.
- Use of facial images raises biometric privacy considerations requiring appropriate policy and consent.
- Dependent on the availability of multiple AWS services; an outage in any one can affect functionality.

18. Future Enhancements

- Better anti-spoofing / liveness detection to prevent attendance fraud via photos or videos.
- A dedicated native mobile application, in addition to mobile browser support.
- More advanced analytics and attendance trend dashboards.
- Notifications (email/SMS alerts for absences or late arrivals).
- Multi-location support for organizations with multiple offices/sites.
- Employee self-service features (viewing personal history, requesting corrections).

19. Conclusion

The Serverless Attendance System with Face Recognition demonstrates how a fully serverless, event-driven architecture on AWS can build a contactless, automated employee attendance solution. By combining API Gateway, Lambda, S3, Rekognition, DynamoDB, and EventBridge, the system captures, recognizes, records, and reports employee attendance with minimal manual intervention, while security is addressed through least-privilege IAM roles, encryption, and short-lived storage of biometric data. This architecture is a practical example of applying managed cloud services to a real-world operational problem without the burden of server management.